

COMP 10280 Programming I (Conversion)

Practical Sheet 3

Dé hAoine, 18 Meán Fómhair 2026

Friday, 18 September 2026

For each of the following programs, use assignment statements to give values to variables. When asked to use a float, use the first six digits of your student number as the amount before the decimal point and the last two digits as the amount after the decimal point.

1. Write a program that takes an amount of currency (a float) and an exchange rate to another currency (a float) and prints out the value of the original amount in the other currency. (Use today's exchange rate for two currencies of your choice.) Save this program as p3p1.py.
2. Write a program that takes a single length (a float) and calculates the following:
 - The area of a square with side of that length
 - The volume of a cube with side of that length
 - The area of a circle with diameter of that length
 - The volume of a sphere with diameter of that length
 - The volume of a cylinder with diameter of that length and side of that length

You can use 3.1415927 for the value pi. Save this program as p3p2.py.

3. Write a program that takes an amount (a float), divides the amount in the ratio 60:40, calculates the tax due according to two different tax rates (13.5% on the larger amount and 23% on the smaller) and prints out the total amount (initial amount plus taxes). Save this program as p3p3.py.

**Please upload your work to
the Brightspace site before Sunday
evening.**

**You should keep a copy of your programs
for your portfolio.**